Network Address Translation (NAT)

Project Overview

This project was initiated to explore the functionality and role of Network Address Translation (NAT) in modern IP networks. The main goal is to understand how NAT enables IP address conservation, security, and communication between private and public networks. Key objectives include:

- Understand the basic NAT process, where private IP addresses are translated to public IP addresses and vice versa, enabling devices in a private network to access external networks.
- Review the three main types of NAT: Static NAT (one-to-one mapping for servers), Dynamic NAT (one-to-many mapping using a pool of public IPs), and Port Address Translation (PAT) (many-to-one mapping using ports).
- Analyze NAT configuration concepts such as inside local, inside global, and outside local addresses, and how translation tables map IPs and ports.
- Identify affected network locations including edge routers, internal private networks, DMZ/server zones, VPN endpoints, and multi-homed networks.
- Evaluate the advantages of NAT like IP address conservation, enhanced security and privacy, flexibility, scalability, and multi-homing support.
- Assess potential drawbacks including resource consumption, added latency, compatibility issues with some protocols, and complications with tunneling and VPNs.
- Apply best practices such as choosing the appropriate NAT type, maintaining security, monitoring performance, documenting mappings, and planning for NAT traversal.

Description

Network Address Translation (NAT) is a process that enables one, unique IP address to represent an entire group of computers. In network address translation, a network device, often a router or NAT firewall, assigns a computer or computers inside a private network a public address.

Recommendations

- Use NAT to conserve IPv4 addresses
- Choose the appropriate NAT type
- Maintain proper security
- Avoid NAT with IPv6
- Monitor NAT performance
- Plan for NAT traversal
- Document NAT mappings
- Test thoroughly

References

https://www.geeksforgeeks.org/computer-networks/network-address-translation-nat/
https://www.cisco.com/site/us/en/learn/topics/networking/what-is-network-address-translation-nat.html
https://www.fortinet.com/uk/resources/cyberglossary/network-address-translation
https://www.vmware.com/topics/network-address-translation

Details

NAT is typically implemented on a router, a device that connects two networks. When a device on the private network sends data to a device on the public network, the router intercepts the data and replaces the source IP address with its own public IP address. The router then sends the data to the destination device.

When the destination device sends data back to the router, the router intercepts this data and replaces the public IP address with the original source IP address. The router then sends the data to the original source device. This process is transparent to the devices on both networks.

Generally, the border router is configured for NAT i.e. the router which has one interface in the local (inside) network and one interface in the global (outside) network. When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address. When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.

If NAT runs out of addresses, i.e., no address is left in the pool configured then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent.

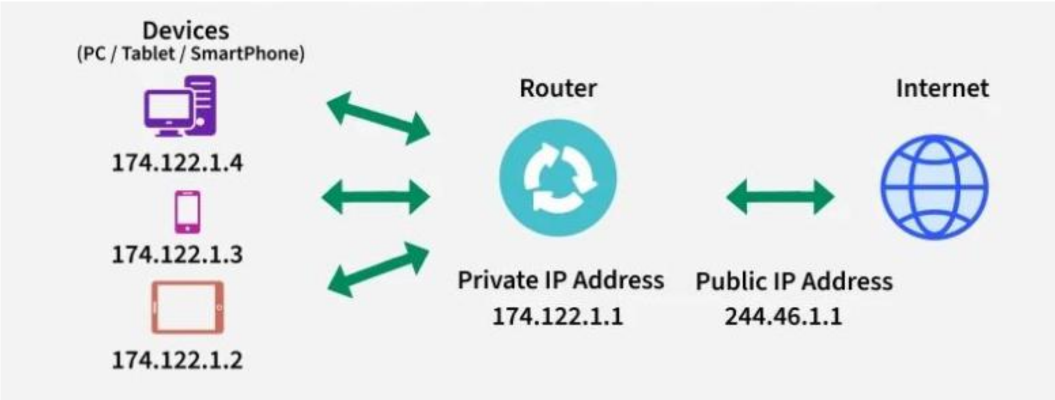

Figure 1: Network Address Translation

Three Types of Network Address Translation (NAT):

1. Static NAT:

   Maps each internal IP to a unique external IP (one-to-one). Used for servers (e.g., web, email) that must be accessible from the internet. The router replaces IPs in both outgoing and incoming traffic accordingly.

2. Dynamic NAT:

   Maps internal IPs to a pool of external IPs (one-to-many). Used mainly for outbound internet access. The router assigns a free external IP from the pool for outgoing traffic and reverses it for incoming traffic.

3. Port Address Translation (PAT):

   Maps multiple internal IPs to a single external IP using different port numbers (many-to-one). Common in home networks, it allows multiple devices to share one public IP by differentiating sessions via ports.

NAT Configuration:

A NAT router requires at least two interfaces: one for the inside (private network) and one for the outside (public network), plus rules to translate IP addresses between them.

- Inside Local Addresses: Private, unregistered IPs used within the internal network (e.g., 192.168.x.x).
- Inside Global Addresses: Registered public IPs assigned by the ISP, used to represent internal devices externally.
- Outside Local Addresses: Used by NAT routers to translate outside global addresses on the public network.

When an internal device sends data outside, the NAT router translates its private IP to a public IP and tracks this in a translation table. Incoming packets from the public network are mapped back to the correct internal device based on this table.

Dynamic NAT & NAT Overloading

- *Dynamic NAT:* Maps private IPs to a pool of public IPs dynamically.
- *NAT Overloading (PAT):* Maps many private IPs to one public IP using different port numbers to distinguish connections.

The NAT router tracks IP and port mappings in a table and translates packets accordingly. When communication ends, entries are removed.

Advantages of NAT

- *Address Conservation*: Saves public IP addresses and prevents depletion.
- *Security & Privacy:* Hides internal IPs from the public network, offering basic firewall protection, traffic filtering, and rate-limiting to help mitigate attacks.
- *Flexibility:* Supports inbound connections with static NAT and works well in various environments like public WLANs.
- *Simplicity:* Eliminates the need to renumber IPs during network changes or mergers.
- *Speed:* Transparent to endpoints and faster than proxy servers since NAT operates at the network layer.
- *Scalability:* Works well with DHCP to easily add devices without requesting more public IPs.
- *Multi-homing Support:* Enables reliable connections and load balancing across multiple ISPs using BGP routing.

Disadvantages of NAT

- *Resource Use:* Consumes router memory and CPU to track and translate IPs.
- *Delays:* Adds processing delay due to address translation.
- *Functionality Issues:* Some applications/protocols may not work properly behind NAT.
- *Traceability:* Complicates tunneling and VPN protocols; IPsec is recommended for secure NAT traversal.
- *Layer Violation:* NAT modifies transport layer info (ports) though routers operate at the network layer, which can cause complications.

Affected Locations

- Edge Routers (NAT Devices)
- Internal (Private) Networks
- External (Public) Networks
- DMZ or Server Zones
- VPN/Tunneling Endpoints
- Multi-homed Networks